

Chromosomal instability & immune escape in driver-negative LUAD

A cleaned record of the session: each prompt, the Asta tools it ran, the results, and the artifacts each step produced.

● PROMPT

Install the Asta plugins from github.com/allenai/asta-plugins and log me in.

Asta CLI · setup + auth

```
claude plugin marketplace add allenai/asta-plugins
claude plugin install asta-tools
asta auth login    # Auth0 device flow → verification link
```

Marketplace added, **asta-tools** installed, Asta CLI **v0.101.1** on PATH, and authentication completed via the device-flow link.

● PROMPT

Here's the data I'll give AutoDiscovery (a Google Drive folder). Download and inspect it before the literature search.

Shell · download + inspect

```
gdown --folder <drive-url>          # TCGA-LUAD multi-omics
python # inspect CNA, expression, mutations, TMB, ssGSEA, RPA classification
```

TCGA-LUAD multi-omics: copy number, RNA-seq, mutations, TMB, Hallmark ssGSEA, and an RPA (RTK/RAS/RAF Pathway Alteration) classification giving **118 driver-negative / 383 driver-positive** cases. TIGIT and LAG3 confirmed present in the expression matrix.

● PROMPT

Run a literature search on driver-negative lung cancer. The Theorizer flagged chromosomal instability driving the tumors, with immune escape — I care about TIGIT and LAG3 specifically, but also identify broader gaps I can explore with AutoDiscovery.

Asta Paper Finder · asta literature find

```
asta literature find "Biology, genomic drivers, and immune microenvironment of
  oncogene-negative (RTK/RAS/RAF wild-type) LUAD; mechanisms of immune evasion." --mod
asta literature find "Chromosomal instability / aneuploidy vs TIGIT and LAG3
  in lung adenocarcinoma." --mode diligent
# 8 searches total (diligent + fast)
```

615 unique papers. Three mature literatures — CIN → immune escape, TIGIT/LAG3, driver-negative LUAD — run in near-isolation: 0 papers link CIN to driver-negative disease; only 2 (generic pan-cancer) touch CIN and TIGIT/LAG3 together. The junction is a genuine white space, framed as five testable gaps.

ARTIFACTS

■ GAP_ANALYSIS.md

● PROMPT

Set up 5 AutoDiscovery runs across the gaps we identified – 20 experiments each.

Asta AutoDiscovery · asta autodiscovery

```
asta autodiscovery create    # one run per gap
asta autodiscovery upload <run> master_sample_table.csv
asta autodiscovery metadata <run> --file run.json    # intent steers exploration
asta autodiscovery submit <run> -y                  # 20 experiments
```

6 runs · 120 experiments. The hypothesis was **refuted in its stated direction**: CIN tracks an immune-*cold* state (checkpoint-low), not driver-specific, not TIGIT/LAG3-selective, not cGAS-STING-mediated. STK11 loss emerged as a focal contributor to coldness.

ARTIFACTS

■ SYNTHESIS.md

■ RUNS.md

● PROMPT

Set up the infiltration-normalized follow-up – after accounting for how many T cells are present, does CIN affect LAG3 *per T cell*?

Asta AutoDiscovery · run 6

```
# adds a CD8 / T-cell infiltration score to the table, then
asta autodiscovery create · upload · metadata · submit    # 20 experiments
```

A dissociation: CIN → fewer T cells, but per infiltrating T cell **LAG3 rose** ($\beta = 2.81$, $p = 0.001$), **TIGIT was null**, IFN- γ rose. A LAG3-specific lead worth stress-testing.

ARTIFACTS

■ SYNTHESIS.md · run 6

● PROMPT

Do a thorough evidence-gathering pass for the best public data to test the LAG3 finding, then design up to 10 Data Voyager experiments.

Evidence gathering + Paper Finder grounding

```
# parallel dataset sweeps (single-cell, bulk cohorts, ICI/protein, TCGA references)
asta literature find "single-cell CIN → CD8 / LAG3 in LUAD" ; "LAG3 biology: FGL1, MHC
```

A ranked catalog of public datasets (single-cell atlases, bulk multi-omic cohorts, ICI-response and protein/spatial sets) plus gold-standard TCGA references, and a tiered 10-experiment plan. Key structural finding: no open dataset carries LAG3 protein + T-cell localization + CIN genomics together, so validation must be layered.

ARTIFACTS

■ EVIDENCE_CATALOG.md

■ DATAVOYAGER_EXPERIMENTS.md

● PROMPT

Assemble the gold-standard datasets, then execute the first tier of Data Voyager experiments.

Asta DataVoyager · asta analyze-data (E1, E2)

```
asta analyze-data submit "In driver-negative LUAD, re-test CIN → LAG3 per T cell using
the Taylor Aneuploidy Score and an independent CD8 estimate, adjusting for tumor pur
WGD and TMB. Report LAG3, TIGIT, IFN-γ; test purity as a confounder." e1.csv
asta analyze-data submit "Repeat across 6 CD8 deconvolution methods (robustness)." e2.
```

Under gold-standard metrics the per-cell LAG3 effect **vanished** ($\beta \approx -0.03$, $p = 0.85$), and CIN ↔ infiltration disappeared ($r = -0.03$) — the run-6 structure had hinged on a crude CIN proxy. Across 7 methods, LAG3 was 0/7 significant.

ARTIFACTS

■ TIER1_RESULTS.md

● PROMPT

Run E1b — pin down whether it's tumor purity or the CIN metric.

Asta DataVoyager · asta analyze-data (E1b)

```
asta analyze-data submit "For each CIN metric (amplitude proxy, fraction-genome-altered, Taylor Aneuploidy Score), regress LAG3 on CIN adjusting for infiltration and TMB — without purity. Report subgroup slopes; TIGIT as comparator." table.csv
```

The "CIN → less infiltration" arm was largely a **tumor-purity artifact** (proxy $r = 0.54$ with purity). The per-cell LAG3 signal was purity-robust but **covariate-dependent** — significant only when adjusting for an expression-derived T-cell score collinear with LAG3.

ARTIFACTS

■ E1B_RESULTS.md

● PROMPT

Pause driver-negative single-cell. Audit the LuCA atlas for eligible tumors, then use a formal CIN × driver-status interaction on TCGA to test whether the effect is truly driver-specific; if it's weak, run the general all-LUAD single-cell analysis and keep driver-negative as a subgroup note.

Asta DataVoyager + LuCA audit + single-cell pipeline

```
# feasibility audit on the LuCA atlas obs (h5py)
asta analyze-data submit "Formal CIN × driver-status interaction on LAG3, both CIN met
  both infiltration covariates; subgroup slopes; TIGIT comparator." table.csv
# interaction weak → general all-LUAD single cell (outside Asta): inferCNV → CD8 pseud
```

LuCA audit: **43 eligible primary LUAD tumors** for a general test, but only **1** eligible onco-negative tumor — driver-negative single-cell is infeasible. The TCGA interaction was **weak/absent** — the effect is general, not driver-specific. In single cell, measuring CD8 LAG3 directly across 43 tumors: $\beta = 0.03$, $p = 0.56$ — null. The lead did not replicate.

ARTIFACTS

■ SC_FEASIBILITY_AUDIT.md

■ E7_INTERACTION_RESULTS.md

■ LUCA_GENERAL_RESULTS.md

● PROMPT

Write an overall project summary from the perspective of a researcher in the field.

Synthesis

A full write-up: motivation, the Asta agents used, every avenue tested, and the scientific + methodological conclusions. Bottom line — the CIN → TIGIT/LAG3 immune-escape hypothesis is not supported; driver-negative LUAD is not a distinct CIN-immune compartment; the durable takeaways are methodological (CIN-metric choice, tumor purity, infiltration modeling) plus STK11-loss coldness as a better-supported lead.

ARTIFACTS

■ PROJECT_SUMMARY.md

Driver-Negative LUAD — Literature Gap Analysis (Final)

Hypothesis under test: In oncogene/driver-negative lung adenocarcinoma (LUAD), chromosomal instability (CIN) drives the tumors and mediates immune escape, with **TIGIT and LAG3** as checkpoints of interest (panel-aware vs PD-1/PD-L1/TIM-3/CTLA-4).

Method: 4 Asta Paper Finder searches ([asta literature find](#)), **all diligent mode**. **615 unique papers**, ~74% published 2021+. (q1/q2 were first run in fast mode when diligent hit transient gateway upstream timeouts; both were later completed in diligent mode on retry — numbers below reflect the deeper diligent corpus.)

ID	FOCUS	MODE	PAPERS
q1	Driver-negative LUAD landscape & immune evasion	diligent	104
q2	CIN → immune evasion mechanism (cGAS-STING/IFN/antigen)	diligent	281
q3	TIGIT/LAG3 in NSCLC vs full checkpoint panel	diligent	130
q4	CIN/aneuploidy ↔ checkpoint expression in LUAD	diligent	129

What is ESTABLISHED (not gaps)

1. **CIN → immune escape is a mature, mechanistic field — but almost entirely pan-cancer / general, not driver-stratified LUAD.** Anchor papers: - *Non-cell-autonomous cancer progression from chromosomal instability* (Nature 2023): CIN → chronic **cGAS-STING** activation → metastasis and immune modulation. - *Genomic instability as a driver and suppressor of anti-tumor immunity* (Front Immunol 2024): the **double-edged sword** — acute CIN is immunogenic, chronic CIN drives evasion. - *Suppression of tumor antigen presentation during aneuploid tumor evolution* (Oncoimmunology 2019): CIN-high tumors are less immunogenic / less infiltrated.
2. **TIGIT and LAG3 as NSCLC checkpoints/targets is well-covered** (TIGIT n=62, LAG3 n=70), including reviews, prognostic meta-analyses, and trial-oriented pieces — but as PD-1-adjacent targets, **not tied to genomic instability**.
3. **Driver/oncogene-negative LUAD is a recognized but thin niche** (15 of 615 papers). The biology anchors: - *Combinatorial inactivation of tumor suppressors efficiently initiates lung adenocarcinoma...* (Cancer Research 2021/2022): "a large fraction of lung adenocarcinomas lacks mutations in known oncogenes... these oncogene-negative tumors remain enigmatic" — and they arise from **tumor-suppressor loss** (TP53/RB1/etc.), itself a canonical CIN driver. - *Lung adenocarcinomas without driver genes converge to common adaptive strategies* (2024). - *Fully accessible fitness landscape of oncogene-negative lung adenocarcinoma* (2023). These characterize genesis/evolution but say nothing about the immune axis or checkpoints.

The GAP (intersection tests across all 615 papers)

JUNCTION	# PAPERS	NOTE
cGAS-STING / micronuclei	267	very large, general
PD-1 / PD-L1	170	very large
Chromosomal instability / aneuploidy	108	large, general
LAG3	70	large, NSCLC
TIGIT	63	large, NSCLC
Oncogene/driver-negative lung	15	thin
CIN AND (TIGIT or LAG3)	2	both generic pan-cancer bioinformatics
CIN AND driver-negative	1	closest prior work (see below) – no immune/checkpoint axis
Driver-negative AND (TIGIT/LAG3)	0	—

Robust to search depth. Doubling the corpus (317 → 615) exploded the *general* mechanism literature – cGAS-STING went 64 → 267 – while the hypothesis junctions barely moved (CIN∩TIGIT/LAG3 stayed at 2; driver-neg∩TIGIT/LAG3 stayed at 0). The white space is not a sampling artifact.

Closest prior work (the single CIN∩driver-negative hit): *Proteogenomic characterization identifies clinical subgroups in EGFR and ALK wild-type never-smoker lung adenocarcinoma* (2024) – profiles wild-type LUAD including copy-number/CIN features, but does **not** address immune escape, TIGIT, or LAG3. It is the nearest neighbor to your hypothesis and the natural citation to differentiate against.

Interpretation. Three mature literatures – CIN → immune-escape, TIGIT/LAG3, and (nascent) driver-negative LUAD – run in near-total isolation. The specific, mechanistically motivated chain you propose sits in a **genuine white space**:

Tumor-suppressor-loss-driven (driver-negative) LUAD is *expected* to be CIN-high; CIN drives immune escape via chronic cGAS–STING; yet **no study tests whether this selectively up-regulates TIGIT/LAG3 in the driver-negative subset**. Zero papers close that loop.

Recommended AutoDiscovery scope (ranked)

Primary (highest novelty, data purpose-built):

In driver-negative LUAD, does chromosomal-instability burden correlate with — and potentially drive — TIGIT/LAG3 up-regulation, more strongly than in driver-positive LUAD and more selectively than for PD-1/PD-L1?

Testable end-to-end in the TCGA data: - **CIN metric** ← `cleaned_cna (1).csv` (fraction genome altered / arm-level aneuploidy). - **Driver status** ← `ON classification TCGA.xlsx` (`Previous_RPA_Positive`: 0 = driver-neg, n=118; 1 = driver-pos, n=383). - **Checkpoints** ← `cleaned_gene_expression.csv.gz` (TIGIT ENSG00000181847, LAG3 ENSG00000089692, + PD-1/PD-L1/TIM-3/CTLA-4). - **Mechanism readout** ← ssGSEA `HALLMARK_INTERFERON_GAMMA/ALPHA_RESPONSE`, `INFLAMMATORY_RESPONSE` (proxy for the cGAS-STING/IFN axis). - **Confounder** ← `cleaned_tmb.csv` (TMB co-varies with both CIN and immunity — must adjust).

Secondary gaps worth an AutoDS arm each: 1. Is the CIN → checkpoint effect *specific* to driver-negative tumors? (interaction test) 2. Does CIN elevate TIGIT/LAG3 *selectively* vs PD-1/PD-L1? (differential axis) 3. Is the CIN → checkpoint link *mediated* by the IFN/cGAS-STING hallmark signature? (mediation, using the ssGSEA scores)

Caveats

- All four searches are now **diligent mode** (615 papers). q1/q2 initially fell back to fast mode when diligent hit transient gateway upstream timeouts, but both completed on retry; the deeper corpus only strengthened the gap (intersections stayed at 0–2 while the general literature roughly doubled).
 - "Driver-negative" here = RTK/RAS/RAF-pathway wild-type per the TCGA RPA classification; other definitions (e.g. including STK11/KEAP1 status) could shift the subset.
 - Intersection counts are keyword-based on title+abstract, so they undercount papers that discuss a junction only in full text. The *direction* (near-empty intersections against large single-concept pools) is robust, but exact counts are lower bounds.
-

AutoDiscovery Synthesis — Does CIN drive immune escape via TIGIT/LAG3 in driver-negative LUAD?

5 runs · 100 experiments · all SUCCEEDED · TCGA-LUAD (n≈115 driver-negative with complete omics).

Headline

The theorizer's hypothesis is refuted — and in the opposite direction. Across all five runs, chromosomal instability (CIN) in driver-negative LUAD is robustly associated with an immune-**cold** microenvironment: **lower** TIGIT/LAG3 (and other checkpoints) and strongly **suppressed** interferon- γ response — not the predicted CIN → cGAS-STING → checkpoint-high exhaustion. This is consistent with the aneuploidy → immune-exclusion arm of the literature (the "double-edged sword"), not the checkpoint-mediated-escape arm.

⚠ Critical caveat (bulk RNA): TIGIT/LAG3 are expressed mainly on infiltrating T/NK cells. Lower *bulk* checkpoint expression in CIN-high tumors most likely reflects **fewer infiltrating lymphocytes**, not lower exhaustion *per T cell*. The single most important follow-up is to normalize checkpoints to a CD8/T-cell-infiltration signature (or deconvolve) and re-test — that could still rescue a per-cell version of the original hypothesis.

Per-gap verdicts

GAP	RUN	VERDICT	KEY STATISTIC
G1 Association	run1	✗ Refuted (opposite sign) – CIN is <i>negatively</i> associated with TIGIT/LAG3	TIGIT vs CN gain $\beta = -3.23$ ($p = 0.009$), loss $\beta = -2.81$ ($p = 0.005$); no inverted-U (quadratic $p = 0.09/0.11$)
G2 Specificity	run2	✗ Not specific to driver-negative – CIN suppresses checkpoints regardless of driver status	CIN main effect on TIGIT $\beta = -3.33$ ($p < 0.001$); CIN×driver interaction $p = 0.74-0.83$ (ns)
G3 Selectivity	run3	✗ No selectivity – CIN hits TIGIT/LAG3 no differently than PD-1/PD-L1	Wald test TIGIT $p = 0.93$, LAG3 $p = 0.85$, PD-1 $p = 0.96$
G4 Mediation	run4	✗ Not mediated by cGAS-STING/IFN – direct association, mechanism unresolved	CIN → STING1 $\beta = -2.5$ ($p = 3e-10$) & STING1 → TIGIT ($p = 0.03$), but indirect effect (ACME) ns; CXCL9 mediation ns
G5 CN events	run5	✓ Global CIN + STK11 loss drive the cold state; not 9p21	CIN → IFN- γ $\beta = -2008$ ($p < 0.001$); only ~17% via focal dels; STK11 loss significant; CDKN2A/B ns

The cross-run picture

- Direction:** CIN → immune-cold, robustly and repeatedly. The core association (G1), the driver-status contrast (G2), and the mechanism runs (G4) all recover a **strong, significant negative** CIN → checkpoint / CIN → IFN- γ effect.
- Not selective, not driver-specific:** the effect is a general immune-coldness phenomenon, not a TIGIT/LAG3-specific or oncogene-negative-specific one. (This actually *answers* Gaps 2–3 – just not in the hypothesized direction.)
- Mechanism:** cGAS-STING is *suppressed* by CIN (consistent with chronic-CIN STING silencing) and correlates with checkpoints, but does **not** formally mediate the CIN → checkpoint link in this bulk cohort. Left unresolved.
- Genomic drivers (G5):** the cold state is driven by *global* instability, with **STK11 loss** the standout focal contributor – echoing the well-known STK11/KEAP1 "cold tumor" biology. 9p21/CDKN2A loss was **not** a significant driver (contra a common assumption).

What this means for the driver-negative literature gap

The literature gap we identified is real, but the data fills it with a *negative/reversal* result: in driver-negative LUAD, CIN does **not** appear to drive TIGIT/LAG3-mediated immune escape; it tracks with immune exclusion. That is still a novel, publishable direction – and it sharpens the next question.

Recommended next AutoDS runs

1. **Infiltration-normalized re-test (highest priority)**: regress TIGIT/LAG3 on CIN *after* adjusting for a CD8/T-cell or immune-infiltration score (e.g. a cytolytic/ `ss_ALLOGRAFT_REJECTION` proxy, or add a deconvolution estimate). Tests exhaustion *per T cell*.
2. **STK11-loss cold program**: characterize the STK11-driven immunosuppressive program in driver-negative tumors specifically.
3. **CIN-high + immune-hot outliers**: identify the driver-negative tumors that are CIN-high yet checkpoint-high (residual analysis) – the exceptions may reveal the escape route.

Run IDs and configs in `RUNS.md`; per-experiment JSON in `runs/exp_<runid>.json`.

Run 6 – Infiltration-normalized re-test (the decisive follow-up)

Question: once you adjust for how many T cells are actually present (`tcell_infiltration_score`), does CIN still predict TIGIT/LAG3 – a *per-cell* effect – or was the negative bulk signal entirely reduced infiltration?

Result – a dissociation, and a partial (LAG3-specific) rescue of the hypothesis:

1. **CIN strongly reduces T-cell infiltration** – confirmed and large: CIN → infiltration $\beta = -4.10$, $p < 0.0001$. This is the dominant effect and explains most of the negative *bulk* CIN → checkpoint association from runs 1–5. (The reduction is **not** mediated by CXCL9 – Path A n.s.)
2. **Per infiltrating T cell, the axes diverge (adjusting for infiltration + TMB)**: - **LAG3: significant POSITIVE** with CIN – $\beta = +2.81$, $p = 0.001$. ← per-cell up-regulation. - **TIGIT: null** – $\beta = 0.03$, $p = 0.95$. - **PD-L1: null** – $\beta = -0.34$, $p = 0.75$. - **IFNG: significant POSITIVE** per T cell – $\beta = +3.74$, $p = 0.015$ (the residing T cells are *more*, not less, functionally activated in high-CIN tumors).
3. **Checkpoint/CD8 ratio method (weaker)**: TIGIT/CD8A and LAG3/CD8A vs CIN were negative but non-significant ($p \approx 0.12$, TMB-adjusted only) – not concordant with the stronger covariate-adjusted regression; treat the per-cell LAG3 signal as suggestive, not settled.

Verdict: The theorizer's hypothesis is **rescued in a refined, LAG3-specific form**. The net bulk story is immune *exclusion* (CIN → fewer T cells), but the T cells that remain show a **per-cell LAG3 up-regulation and elevated IFN- γ** as CIN rises – TIGIT and PD-L1 show no per-cell effect. So CIN-high driver-negative tumors are simultaneously **colder (fewer T cells) yet carry a more LAG3-marked / activated residual T-cell compartment**.

Implication: LAG3 – not TIGIT – is the checkpoint that behaves as the original hypothesis predicted, but only after correcting for infiltration. This is the highest-novelty finding of the study and the cleanest lead for follow-up (ideally single-cell / deconvolution to confirm the per-cell interpretation, since regression-adjustment and ratio methods disagree at the margin).

AutoDiscovery Runs — Driver-Negative LUAD: CIN → immune escape (TIGIT/LAG3)

Hosted AutoDiscovery runs testing the theorizer’s hypothesis. All 5 submitted 2026-07-23, **20 experiments each** (100 credits total), on the shared analysis-ready table `derived/master_sample_table.csv` (501 TCGA-LUAD samples; 118 driver-negative / 383 driver-positive, RPA wild-type definition).

RUN	GAP	HYPOTHESIS / INTENT	RUN ID
1 association	G1	CIN burden ↔ TIGIT/LAG3 within driver-negative LUAD	<code>c9508770-af22-4441-aff0-ac028c2e831d</code>
2 specificity	G2	CIN×driver-status interaction — is the effect specific to driver-negative?	<code>34d6418f-0341-45b6-8d5f-93284984d592</code>
3 selectivity	G3	CIN elevates TIGIT/LAG3 selectively vs PD-1/PD-L1/TIM-3/CTLA-4?	<code>14653a15-4d45-4450-8a90-b62a2c61d61f</code>
4 mechanism	G4	Is CIN → TIGIT/LAG3 mediated by IFN/cGAS-STING?	<code>8ceb95fa-2cf1-413c-806c-aaef41111044</code>
5 CN events	G5	Which specific CN events (9p21/B2M/TP53/RB1/...) predict checkpoint-high?	<code>d9363c9e-b5b9-4b21-afc1-35045f79ba44</code>
6 infiltration-norm	follow-up	Does CIN affect TIGIT/LAG3 <i>per infiltrating T cell</i> (adjusting for <code>tcell_infiltration_score</code>)? Uses <code>master_sample_table_v2.csv</code>	<code>cd05dc30-0443-4a91-abef-67b50beb999d</code>

Run 6 context: the 5-run study found CIN negatively associates with *bulk* TIGIT/LAG3. Since TIGIT tracks T-cell infiltration ($r=0.92$) and CIN reduces infiltration ($r=-0.50$), run 6 adds a CD8/T-cell infiltration score + cytolytic score and asks whether a residual (per-cell) CIN → checkpoint effect survives infiltration adjustment.

Config: runs 1–4 `exploration_weight=2.0, surprisal_width=0.4, evidence_weight=2.5`; run 5 (exploratory) `4.0 / 0.3 / 2.0`. All `mcts_selection=ucb1_recursive`. Every run’s intent instructs adjusting for **TMB** (confounder).

Monitoring

```
export PATH="$HOME/.local/bin:$PATH"  
asta autodiscovery status <runid>  
asta autodiscovery experiments <runid> --format text  
asta autodiscovery run <runid> --format text
```

Evidence Catalog — public data to test the per-cell LAG3 finding

Finding to validate: In oncogene/driver-negative (RTK/RAS/RAF wild-type) LUAD, chromosomal instability (CIN) is associated with (a) **reduced** T-cell infiltration but (b) **increased LAG3 (not TIGIT) per infiltrating T cell**, with elevated per-cell IFN- γ . Derived from TCGA bulk RNA with crude proxies; needs (1) proxy-hardening, (2) composition-free confirmation, (3) independent replication, (4) protein/clinical relevance.

Key structural conclusion from the search

No single open dataset has LAG3 protein + T-cell localization + genomics-for-CIN in the same NSCLC samples. So the validation strategy is layered: harden in TCGA → confirm per-cell in single-cell → replicate in open bulk cohorts → corroborate at protein/clinical level (some access-gated). Notably, the closest protein evidence (**Datar & Schalper 2019, CCR**) independently found that **baseline LAG-3 in TILs predicts anti-PD-1 non-response while PD-1/TIM-3 do not** — a protein-level echo of our LAG3-specific, per-T-cell hypothesis.

Tier A — Harden the TCGA analysis (open, one-click GDC downloads)

Replace our two proxies and add the critical purity confounder.

RESOURCE	WHAT IT GIVES	DOWNLOAD	REPLACE
Thorsson 2018 leukocyte fraction	Methylation-based immune infiltrate per TCGA sample (RNA-independent)	GDC panimmune <code>TCGA_all_leuk_estimate.masked.20170107.tsv</code>	our self-computed score
Thorsson CIBERSORT fractions	22 immune subsets incl. CD8 (× leukocyte fraction = absolute CD8)	GDC panimmune <code>TCGA.Kallisto...cibersort.relative.tsv</code>	independent abundance
Taylor 2018 Aneuploidy Score	Purity-aware per-sample AS (0–39) + arm-level calls	GDC pancan-aneuploidy <code>PANCAN_ArmCallsAndAneuploidyScore_092817.txt</code>	our FG, not a score
ABSOLUTE purity/ploidy/WGD	Tumor purity, ploidy, genome-doublings	GDC pancanatlas <code>TCGA_mastercalls.abs_tables_JSedit.fixed.txt</code>	purity confounder (purity bulk immune signal) 2nd CII
TIMER2.0 / immunedeconv	6-method CD8 estimates	timer.cistrome.org / omnideconv/immunedeconv	multi-robust
CIBERSORTx high-res / CODEFACS	cell-type-specific LAG3 from bulk	cibersortx.stanford.edu (token) / ruppilab	turns "LAG3" into inferred measure
Saltz 2018 TIL maps	H&E image-based TIL fraction (orthogonal)	GDC tilmap / TCIA	image confirmed infiltration

Tier B — Single-cell LUAD (open; composition-free per-cell test via inferCNV/CopyKAT)

DATASET	N	ACCESS	NOTE
Salcher 2022 LuCA (best-powered)	1.3M cells / 318 pts, LUAD- dominant	OPEN — CELLxGENE (h5ad/rds) + Zenodo	only atlas where driver-negative is a viable subgroup; genotype for a subset; batch/integration caveat
Kim 2020 GSE131907	208k cells / 44 pts	OPEN — GEO	largest single consistent 10x LUAD; malignant + deep T; driver annotation partial
Bischoff 2021	114k cells / 10 LUAD	OPEN — Code Ocean	most complete per specimen (deep CD8 + malignant + genotype); small n
Maynard 2020 PRJNA591860	23k cells / 30 pts, Smart- seq2	OPEN (processed, Code Ocean)	truest per-cell reads + clear driver status; treatment-exposed
Laughney 2020 GSE123902 / Lambrechts E-MTAB- 6149/6653	small	OPEN	supporting replication

Tier C — Independent bulk replication (CNA+RNA+mutations, same tumors)

COHORT	N LUAD	ACCESS	CIN?	DRIVER-NEG?
OncoSG luad_oncosg_2020	305 (East- Asian)	OPEN — cBioPortal	✓ GISTIC	✓
CPTAC-LUAD luad_cptac_2020	110 (+ NAT)	OPEN — cBioPortal; +proteomics	✓	✓
TRACERx421	~248 LUAD, multi-region	CIN/driver OPEN (suppl.); RNA EGA- controlled	✓ (open tables: wGII/WGD)	✓
Korean EGFR/ALK-WT never-smoker	99, oncogene- neg <i>by design</i>	Controlled (KoNA); proteomics open (PXD033360)	✓	✓ exact- match
Sherlock-Lung	232 never- smoker	RNA open GSE171415; WGS dbGaP	✓ (controlled)	✓

Tier D — Immunotherapy response & LAG3 protein/spatial

DATASET	WHAT	ACCESS
Ravi 2023 / SU2C-MARK	393 NSCLC ICI; WES+RNA+response+survival – supports the <i>entire</i> hypothesis	Controlled – dbGaP phs002822
Rizvi 2015 luad_mskcc_2015 / Hellmann 2018 nsc1c_mskcc_2018	WES + ICI response (CIN/TMB derivable; no RNA)	OPEN – cBioPortal
GSE126044 / GSE135222	anti-PD-1 NSCLC bulk RNA + response (no CNA)	OPEN – GEO
Datar & Schalper 2019	QIF LAG-3/PD-1/TIM-3 protein per TIL, >800 NSCLC + 90 ICI – <i>the</i> on-point protein biology	By request / MTA (Yale)
Sorin 2023 Nature IMC	416 LUAD, 35-plex, 1.6M cells + survival – T-cell/tumor spatial (no LAG3 channel, no CIN)	OPEN – Zenodo 7383627
Walsh/Quail 2025 IMC	102 NSCLC spatial (no LAG3, no CIN)	OPEN – Zenodo 14625562

Clinical fact base: RELATIVITY-104 (nivolumab+relatlimab 1L NSCLC; ORR 51.3% vs 43.7%, benefit strongest in PD-L1 $\geq$ 1% non-squamous) and TACTI-002 (eftilagimod+pembrolizumab) establish LAG-3 as an actionable co-checkpoint in NSCLC – raising the stakes of a CIN → LAG3 axis.

Data Voyager experiment plan — understanding the LAG3 effect (10 experiments)

Each experiment = (dataset, research question) for the DataVoyager `analyze-data` agent, which runs code on a local tabular file. "Prep" = what must be assembled into an analysis-ready table first; "Runnable now" = data already local or one-click open.

Central claim under test: **in driver-negative LUAD, CIN → fewer T cells but higher LAG3 (not TIGIT) & IFN-γ per T cell**. The plan moves: harden → confirm per-cell → replicate → mechanism → clinical.

Tier 1 — Harden the TCGA finding (runnable now; kills the biggest confounds)

E1. Gold-standard re-test. *In driver-negative TCGA-LUAD, does per-T-cell LAG3 still rise with aneuploidy after replacing both proxies and adjusting for tumor purity?* - Data: our `master_sample_table` + Taylor **Aneuploidy Score**, Thorsson **leukocyte fraction** & CIBERSORT **absolute CD8**, **ABSOLUTE purity/ploidy/WGD**. Prep: join GDC tables on barcode. - Discriminates: whether the effect survives purity adjustment (purity dilutes bulk immune signal — the untested confound most likely to explain it away) and better metrics. **Highest priority.**

E2. Infiltration-method robustness. *Is the per-cell LAG3~CIN signal stable across 6 deconvolution methods (CIBERSORT-ABS, quanTIseq, xCell, MCP-counter, EPIC, TIMER)?* - Data: TIMER2.0 per-sample CD8 estimates (all methods) joined to LAG3/AS. Prep: pull TIMER2.0 LUAD table. - Discriminates: whether the earlier regression-vs-ratio discordance is a method artifact or robust.

E3. Cell-type-specific LAG3 (measured, not inferred). *Estimate LAG3 expression within the CD8/T-cell compartment per sample (CIBERSORTx high-res / CODEFACS) and regress on aneuploidy in driver-negative LUAD.* - Data: TCGA-LUAD bulk RNA → CIBERSORTx high-res GEPs. Prep: run CIBERSORTx (token) with a LUAD scRNA reference; produces per-sample T-cell-specific LAG3. - Discriminates: converts "per-cell LAG3" from a covariate-adjusted inference into a direct measurement.

Tier 2 — Single-cell confirmation (composition-free; the decisive layer)

E4. Per-cell LAG3 vs tumor CNV in scRNA. *Within CD8 T cells, is LAG3 higher in patients whose malignant cells show high inferred aneuploidy?* - Data: **Kim GSE131907** (and/or **LuCA**); inferCNV/CopyKAT on malignant cells → per-tumor CNV burden; pseudobulk/per-cell CD8 LAG3. Prep: scRNA download + inferCNV + build per-cell/per-tumor table. - Discriminates: the definitive composition-free test — bulk cannot separate this.

E5. Reproduce the dissociation at single-cell. *Does CNV-high associate with (a) lower CD8 fraction AND (b) higher per-CD8 LAG3 in the same cohort?* - Data: same scRNA tables as E4. - Discriminates: confirms the two-part dissociation (exclusion + per-cell up-regulation) directly.

E6. Selectivity + activation-vs-exhaustion state. *Is the effect LAG3-selective vs TIGIT/PD-1, and are high-LAG3 CD8 cells activated effectors (IFNG⁺, GZMB⁺) or terminally exhausted (TOX⁺, TCF7⁻)?* - Data: scRNA CD8 subset with checkpoint + exhaustion/effector programs vs per-tumor CNV. - Discriminates: interprets the elevated per-cell IFN-γ – genuine activation vs early/terminal exhaustion; central to the biological meaning.

Tier 3 – Independent bulk replication (open cBioPortal)

E7. OncoSG replication. *Does infiltration-adjusted LAG3 (> TIGIT) ~CIN replicate in OncoSG LUAD?* - Data: [luad_oncosg_2020](#) CNA+RNA+mutations. Prep: cBioPortal download → build master-style table + derive AS + driver-negative + a CD8 signature. - Discriminates: independent East-Asian replication.

E8. CPTAC replication at RNA and protein. *Does it hold in CPTAC-LUAD, including LAG3/CD8 at the protein level?* - Data: [luad_cptac_2020](#) CNA+RNA+**proteomics**. Prep: build table incl. protein LAG3/CD8. - Discriminates: cross-platform + cross-ancestry, and RNA → protein consistency.

Tier 4 – Mechanism & clinical relevance

E9. Functional coherence (ligand + STK11). *Does CIN track with LAG3's functional context – FGL1 and MHC-II expression – and does STK11 loss (our run5 focal hit) modulate the LAG3 axis?* - Data: TCGA + OncoSG (FGL1, HLA-DR/CIITA, CN_STK11, AS, LAG3). - Discriminates: whether a CIN → LAG3 axis is functionally actionable (ligand present) vs epiphenomenal; connects to the STK11-cold result.

E10. Clinical / ICI relevance. *Is high T-cell-normalized LAG3 associated with anti-PD-1 non-response in NSCLC – and does the CIN → LAG3 axis mark a PD-1-resistant, LAG3-co-blockade-candidate subset?* - Data: **GSE126044 + GSE135222** (open RNA + response) for the LAG3~response test; **Rizvi 2015 / Hellmann 2018** (open WES) for CIN~response; **Ravi/SU2C-MARK** (dbGaP, if approved) for the full CIN×LAG3×response model. Prep: assemble per-patient tables. - Discriminates: whether the finding has therapeutic traction (mirrors Datar/Schalper 2019 protein result).

Sequencing & access summary

- **Run immediately (open + mostly local):** E1, E2, E7, E9, and the open half of E10 (GSE126044/135222, Rizvi/Hellmann). E3 open but needs a CIBERSORTx run.
- **Needs download + preprocessing (open):** E4, E5, E6 (scRNA + inferCNV), E8 (CPTAC protein).

- **Access-gated (parallel applications):** Ravi/SU2C-MARK dbGaP (E10 full model); TRACERx EGA (confirmatory); Datar/Schalper MTA (protein gold standard).
 - **Recommended first batch:** E1 → E3 → E4 (harden → measure → confirm), then E7/E8 replication.
-

Tier-1 DataVoyager results – the per-cell LAG3 finding does NOT survive gold- standard re-testing

Two DataVoyager experiments re-tested the run6 finding (in driver-negative LUAD, CIN reduces T-cell infiltration but raises LAG3 per T cell) after replacing our crude proxies with publication-grade metrics. **The effect largely collapses.**

E1 – Gold-standard re-test (Taylor Aneuploidy Score + absolute CD8 + purity/WGD)

Driver-negative LUAD (n≈115). Standardized predictors; OLS adjusting for infiltration (CIBERSORT absolute CD8), purity, WGD, TMB.

MODEL	ANEUPLOIDY_SCORE COEF	P
LAG3 (with purity)	-0.027	0.85
LAG3 (no purity)	-0.011	0.94
TIGIT	-0.054	0.65
IFNG	+0.209	0.39
aneuploidy ↔ infiltration (Pearson)	r = -0.026	0.79

- The per-cell **LAG3 effect vanishes**: run6 had $\beta = +2.81$, $p = 0.001$; here $\beta \approx -0.03$, $p = 0.85$.
- **IFN- γ per cell** also null (was $+3.74$, $p = 0.015$).
- **Purity is not the culprit** – adding/removing it barely moves LAG3.
- **The most telling result**: with the standard, purity-corrected Aneuploidy Score, CIN has **no association with T-cell infiltration** ($r = -0.03$, $p = 0.79$) – whereas our proxy gave $\beta = -4.10$, $p < 1e-4$. The entire run6 structure hinged on the crude CIN proxy.

E2 – Robustness across 7 infiltration-deconvolution methods

Driver-negative LUAD; LAG3~aneuploidy adjusting for each method's CD8 + purity + TMB.

- **LAG3: 0/7 methods significant**. All 7 coefficients *positive* (0.039–0.173) but all $p > 0.10$.
- **TIGIT: 0/7 significant**. All 7 *negative* (–0.009 to –0.137).

So a **faint, non-significant directional echo persists** (LAG3 positive, TIGIT negative across every method) — the LAG3 > TIGIT pattern isn't random — but it does **not** reach significance.

Verdict

The strong per-cell LAG3 claim is not supported by rigorous metrics. With the citable, purity-aware Aneuploidy Score and independent infiltration estimates, neither the infiltration-reduction nor the per-cell LAG3/IFN- γ up-regulation replicates. The run6 result was substantially an **artifact of the crude CIN proxy** (`cna_mean_abs_log2` / fraction-genome-altered) and/or the self-computed CD8 score — not a robust biological signal.

What survives is weak: a **consistent-but-non-significant** tendency for LAG3 (not TIGIT) to trend positive with aneuploidy per T cell (E2, 7/7 same direction). Not disproven, but far below the confidence run6 implied.

Why the discordance — and the decisive next test

run6's CIN metric (amplitude / fraction-genome-altered) behaves very differently from the arm-level **Aneuploidy Score**. Two explanations, separable with one cheap experiment: - **(a) Purity/proxy artifact**: the FGA proxy is compressed by low tumor purity, which co-varies with stromal/immune content — manufacturing a spurious CIN $\leftrightarrow$ immune link. - **(b) Genuine metric-type biology**: amplitude/focal instability $\neq$ arm-level aneuploidy; the effect could be real for one axis and not the other.

Proposed E1b (run next): re-fit the run6 models using the *original* proxy `cna_mean_abs_log2` as the CIN metric, driver-negative, **with vs without purity adjustment**, alongside Aneuploidy Score in the same table. If purity adjustment kills the proxy-based effect $\rightarrow$ (a) artifact. If it survives with purity but AS still shows nothing $\rightarrow$ (b) metric-type distinction worth pursuing.

Bearing on the plan

- Tiers 2–4 (single-cell, replication, clinical) are **lower priority** until E1b clarifies whether there is any real effect to chase. The single-cell test (E4–E6) remains the true arbiter of a *per-cell* claim, but should be gated on E1b rather than run speculatively.
 - This is the intended function of validation: a proxy-driven finding was caught before it drove expensive downstream work.
-

E1b – disentangling CIN metric, purity, and infiltration covariate

Decisive follow-up to E1. Same driver-negative TCGA-LUAD data, but testing all three CIN metrics (`cna_mean_abs_log2` = run6 amplitude proxy; `cin_frac_altered_0.3` = FGA proxy; `aneuploidy_score` = Taylor arm-level) $\times$ $\pm$ purity, under two infiltration covariates. **The finding splits cleanly into two arms with opposite fates.**

Arm 1 – "CIN reduces T-cell infiltration" (run6 $\beta = -4.10$, $p < 1e-4$) $\rightarrow$ purity artifact

PART B: CIN $\rightarrow$ tcell_infiltration_score, $\pm$ purity, and each metric's correlation with purity:

CIN METRIC	COEF (NO PURITY)	COEF (+ PURITY)	CORR WITH PURITY
cna_mean_abs_log2	-0.501, $p < 0.001$	-0.150, $p = 0.092$ (ns)	$r = 0.54$
cin_frac_altered_0.3	-0.490, $p < 0.001$	-0.167, $p = 0.046$	$r = 0.54$
aneuploidy_score	-0.032, $p = 0.74$ (ns)	-0.137, $p = 0.058$	$r = -0.10$

Verdict: the infiltration-exclusion effect is **substantially purity-driven**. The amplitude/FGA proxies correlate with tumor purity at $r = 0.54$ (high-purity tumors have less stroma/immune); adjusting for purity collapses the effect ($p \rightarrow 0.09$), and the purity-robust Aneuploidy Score ($r = -0.10$ with purity) shows **no infiltration effect at all** ($p = 0.74$). run6's headline exclusion result does not survive.

Arm 2 – "per-cell LAG3 rises with CIN" $\rightarrow$ survives purity, LAG3-specific, but covariate-dependent

PART A (LAG3 $\sim$ CIN + tcell_infiltration_score + tmb):

CIN METRIC	COEF (NO PURITY)	COEF (+ PURITY)
cna_mean_abs_log2	+0.328, $p = 0.001$	+0.364, $p = 0.001$
cin_frac_altered_0.3	+0.227, $p = 0.023$	+0.245, $p = 0.021$
aneuploidy_score	+0.187, $p = 0.036$	+0.201, $p = 0.029$

- **Positive and significant for all three CIN metrics, and purity does NOT abolish it** (unlike Arm 1).
- TIGIT (Part A2): null for every metric ($p > 0.3$) $\rightarrow$ LAG3-specificity holds.

PART C2 sensitivity (LAG3 ~ CIN + **cibersort_CD8_absolute** + tmb) – swapping the infiltration covariate:

CIN METRIC	COEF (NO PURITY)	COEF (+ PURITY)
cna_mean_abs_log2	+0.136, p=0.228	+0.311, p=0.011
cin_frac_altered_0.3	+0.062, p=0.568	+0.201, p=0.083
aneuploidy_score	+0.164, p=0.103	+0.124, p=0.215

Verdict: the per-cell LAG3 signal is real but **fragile and covariate-dependent** – robust and significant when adjusting for the expression-based T-cell score (Part A), but only the amplitude metric (+purity) reaches significance when adjusting for the independent CIBERSORT-absolute estimate (Part C2). This resolves the E1 ↔ run6 discrepancy: E1 used cibersort_CD8_absolute (mostly null); run6/E1b Part A used the expression T-cell score (significant).

⚠ Circularity caveat: `tcell_infiltration_score` is built from CD8A/CD3D/... *expression* in the same RNA sample as LAG3, so adjusting LAG3 for it is expression-on-expression and can leave residual co-expression. The more independent CIBERSORT-absolute estimate gives a weaker signal. This is exactly the bulk-deconvolution ambiguity that **single-cell data removes**.

Metric relationships (Part C1)

cna_mean_abs_log2 ↔ cin_frac_altered_0.3: r=0.90; cna ↔ aneuploidy_score: r=0.61; cin_frac ↔ AS: r=0.52. The two proxies are near-identical; both diverge moderately from arm-level aneuploidy.

Overall verdict & implication for the plan

- **Infiltration-exclusion arm: retract** – a tumor-purity artifact.
 - **Per-cell LAG3 arm: alive but unresolved** – purity-robust and LAG3-specific, but its significance hinges on how infiltration is modeled (bulk deconvolution can't cleanly separate per-cell level from composition). Not the confident result run6 implied, not the dead result E1 alone implied.
 - **This now *justifies* the single-cell tier (E4-E6)** rather than gating it out: single-cell measures per-cell LAG3 in CD8 T cells directly, eliminating the deconvolution/covariate ambiguity that is the sole remaining source of doubt. That is the definitive arbiter.
-

TCGA CIN × driver-status interaction — the gating test

Question: does driver status genuinely *modify* the CIN → LAG3 association (i.e. is the effect driver-negative-specific), or is it a general LUAD phenomenon? Formal interaction model on the full TCGA-LUAD cohort, across both CIN metrics × both infiltration covariates, + subgroup slopes, + TIGIT comparator.

Result: WEAK / ABSENT interaction — the effect is NOT driver-specific

LAG3 interaction term (CIN_z × driver_positive_RPA), all 4 models:

CIN METRIC	INFILTRATION COV	INTERACTION COEF	P
cna_mean_abs_log2	tcell_infiltration	-0.077	0.44
cna_mean_abs_log2	cibersort_CD8_abs	-0.117	0.30
aneuploidy_score	tcell_infiltration	+0.061	0.56
aneuploidy_score	cibersort_CD8_abs	-0.015	0.90

All non-significant (p 0.30–0.90). **No evidence that driver status modifies the slope.**

Subgroup slopes (CIN → LAG3), driver-negative vs driver-positive — they do not differ materially:

MODEL	DRIVER-NEGATIVE B (P)	DRIVER-POSITIVE B (P)
cna × tcell_infil	0.332 (0.001)	0.240 (0.0005)
cna × cibersort	0.283 (0.011)	0.053 (0.48)
aneuploidy × tcell_infil	0.202 (0.029)	0.213 (0.0002)
aneuploidy × cibersort	0.125 (0.22)	0.069 (0.27)

The CIN → LAG3 slope is **positive in BOTH subgroups** (driver-negative numerically a touch larger but not significantly so). The covariate-dependence noted in E1b persists (stronger with the expression T-cell score, weaker with CIBERSORT-abs) — but that is orthogonal to the interaction question.

TIGIT comparator: interaction ns in all models (p 0.32–0.97); subgroup slopes null or only significant (negative) in driver-positive for one covariate → confirms the positive per-cell signal is **LAG3-specific**, not a generic checkpoint effect.

Verdict & branch taken

Driver status does **not** modify the CIN-LAG3 association → the effect is a **general LUAD** phenomenon, present in oncogene-negative and oncogene-positive tumors alike. This mirrors the original AutoDS run2 finding (interaction ns) but now with gold-standard metrics and both covariates.

Consequences (per plan): 1. **Proceed to the general all-LUAD LuCA single-cell analysis** (43 eligible primary tumors) — framed as validation of a *general*/CIN-CD8-LAG3 mechanism, TIGIT as prespecified comparator. 2. **Keep the driver-negative finding as a bulk-only subgroup observation** — real within that subgroup, but not specific to it. 3. **Do NOT pursue the driver-stratified multiplex-IHC route** — that was gated on *strong* evidence of driver-negative specificity, which the interaction test does not provide.

Single-cell feasibility audit — can we validate the DRIVER-NEGATIVE per-cell LAG3 claim?

Verdict: No — not at defensible power with current public single-cell data. LuCA is the only atlas at the right scale, and its driver annotation is too sparse and too soft to support a per-tumor, single-cell test of a *driver-negative*-specific claim. (Counts parsed directly from the atlas's own `patient_metadata_corrected.xlsx`, 303 patients.)

Why (the gate is driver annotation)

- LuCA **does** carry a driver field (`driver_genes` + per-gene EGFR/KRAS/TP53/ALK/BRAF/ERBB2/ROS columns) and **does** separate "malignant cell" from "CD8+ T cell" — so the cell types are there.
- But driver annotation is **sparse**: **75 / 161 LUAD patients**; and the large *treatment-naive primary* LUAD cohorts (Kim 44, Wu 18, Guo 11, ...) have **no** driver annotation at all. Annotated LUAD collapse to ~2 usable cohorts (Maier 24 naive-early; Laughney 14 primary).
- **Explicitly oncogene-negative LUAD ≈ 5–9 tumors**, from 2 cohorts — and "negative" is *inferred from a list*, not a documented wild-type test (absence ≠ tested-negative). Single digits, soft label.
- **No CNV in the atlas** — malignant cells are labeled by expression clustering, so we'd run inferCNV ourselves on expression-defined malignant cells (mild circularity) with no per-cell CN ground truth.
- **The LuCA authors themselves** ran their genotype ↔ CD8 associations on **TCGA bulk (1,026 pts) via Scissor**, not on their single-cell patients — an explicit admission that single-cell driver annotation is too thin for this exact question.

Eligible independent LUAD tumors (tumor = the unit)

FILTER	ESTIMATED N
LUAD + primary + naive + both malignant & CD8 (ignoring driver)	~70–120 (upper band; exact needs an h5ad <code>sample × cell_type</code> crosstab)
...AND driver-annotated	~35–40
...AND driver-negative specifically	~5–9 ← underpowered

What this means for the plan

1. **The driver-negative-specific per-cell claim cannot be validated in single cell right now.** Per directive, we state this plainly rather than pass off general-LUAD single-cell results as if they settled the driver-negative question.
2. **A single-cell test in LUAD *broadly* IS feasible** (~70–120 tumors) and would resolve the one real weakness of the bulk result — the deconvolution/covariate ambiguity from E1b (per-cell LAG3 was purity-robust and LAG3-specific but sensitive to how infiltration was modeled). It just cannot be labeled driver-negative-specific.
3. Where the driver-negative claim therefore stands: on **bulk TCGA (n≈115 driver-negative)** with a per-cell inference that is purity-robust, LAG3-specific, but covariate-dependent — and **no public single-cell cohort can currently adjudicate it for the driver-negative subset.**

Recommended next steps (ranked)

1. **Cheap, decisive first move:** download only the **LuCA core atlas** (or a cell-type subset) and run `sc_lag3_pipeline.py --audit-only` to get the *exact* per-tumor malignant+CD8 counts and the real driver-negative N — confirming the ~5–9 estimate before any compute.
2. **If you want the mechanism nailed (general LUAD):** run the full pipeline on LuCA ignoring driver status (~70–120 tumors) to settle per-cell vs composition for CIN → CD8-LAG3 broadly. Report it as all-LUAD, explicitly not driver-negative.
3. **For the driver-negative claim specifically:** it stays a bulk finding with the E1b caveat. Closing it would require either a driver-annotated scRNA LUAD cohort (not public at scale) or orthogonal validation (multiplex-IHC LAG3 protein per T cell in driver-stratified tumors — e.g. a Datar/Schalper-style TMA, access-gated).
4. Kim remains a **smoke-test/descriptive** cohort only (~13 tumors); the pipeline + smoke test are ready (`pipeline/`), and the Kim full-matrix download is capped in this container.

EXACT counts — LuCA core `--audit-only` (computed directly from the 13 GB h5ad obs)

Confirmed by reading the atlas obs (h5py) and applying eligibility thresholds (≥50 malignant "Tumor cells" & ≥20 "T cell CD8" per sample):

FILTER (LUAD)	N TUMORS
LUAD samples total (156 patients, 411k cells)	277
eligible (≥ 50 malignant & ≥ 20 CD8)	66
eligible & primary	43 ← feasible for a general all-LUAD analysis
eligible & primary & onco-annotated	13
eligible & primary & onco-negative	1 ← driver-negative single-cell INFEASIBLE (confirmed)

The per-sample audit table is `scrna/luca/luca_luad_audit.csv`. This confirms the plan: run the **general all-LUAD** single-cell analysis on ~43 eligible primary LUAD tumors; the driver-negative-specific single-cell test is not possible (n=1).

LuCA general all-LUAD single-cell analysis — the per-cell LAG3 signal does NOT replicate

Definitive single-cell test of the CIN → CD8-LAG3 mechanism in LUAD broadly (not driver-negative; n=1 there). Removes the bulk-deconvolution ambiguity that limited E1b by measuring CD8 LAG3 directly in real CD8 cells and inferring CIN per tumor from malignant cells.

Design (as prespecified; tumor = independent unit)

- **Cohort:** LuCA core, **43 eligible primary LUAD tumors** across **9 datasets** (≥50 malignant & ≥20 CD8 cells/tumor). All-eligible incl. metastases = 66 (sensitivity).
- **CIN:** infercnvpy on 23,843 malignant ("Tumor cells") vs 15,000 normal-epithelial reference cells (AT1/AT2/Club/Ciliated from normal/normal-adjacent); per-tumor CIN = mean CNV magnitude over malignant cells.
- **CD8 readout:** patient/tumor-level **pseudobulk** mean log-expr over 25,232 CD8 T cells.
- **Model:** tumor-level OLS `pb_GENE ~ CIN_z + C(dataset)` (cohort-adjusted); TIGIT prespecified comparator.

Result — null across the board

GENE	COEF (CIN_Z)	P (COHORT-ADJ)	P (NO COHORT ADJ)
LAG3	+0.026	0.56	0.63
TIGIT	-0.016	0.56	0.36
IFNG	-0.033	0.39	0.27

The LAG3 point estimate is **≈0**, not merely non-significant — this is a null, not just underpowered absence of significance. TIGIT (comparator) and IFN-γ likewise null.

Interpretation

The bulk "per-cell LAG3 rises with CIN" signal (run6) **does not survive** when per-cell LAG3 is measured directly in single cell. Combined with the earlier findings, the LAG3 lead fails a graded validation: - run6 (bulk, expression-based infiltration score): +2.81, p=0.001 — the original signal. - E1 (bulk, independent CIBERSORT-abs infiltration): null. - E1b (bulk): purity-robust & LAG3-specific but **covariate-dependent** (significant only with the expression-based score — which is collinear with

LAG3). - E7 interaction: **general, not driver-specific**. - **LuCA single cell (direct CD8 LAG3, per-tumor CIN): null ($p=0.56$, $\text{coef}\approx 0$)**.

The most parsimonious reading: the bulk per-cell signal was largely an artifact of adjusting LAG3 for an expression-derived T-cell score built from the same RNA (residual co-expression), not a true per-cell CIN $\rightarrow$ LAG3 relationship. Single cell — the method that removes that circularity — shows no effect.

Caveats

- $n=43$ tumors limits power for a *small* true effect, but the near-zero point estimate argues against a meaningful missed effect.
- inferCNV CIN is computed on expression-defined malignant cells (mild circularity) and pooled across 9 platforms (cohort-adjusted). A single normal-epithelial reference set was used.
- All-LUAD, not driver-negative (untestable in single cell; $n=1$).

Bottom line for the whole LAG3 arc

No robust, cross-modal evidence supports a CIN $\rightarrow$ CD8-LAG3 mechanism in LUAD. The driver-negative bulk observation stands only as a subgroup finding sensitive to CIN-metric and infiltration-covariate choice, and it does not generalize or replicate at single-cell resolution. Validation did its job: a promising AutoDS lead was stress-tested and ultimately not supported.

Chromosomal instability, immune escape, and LAG3 in oncogene-negative lung adenocarcinoma

An Asta-driven hypothesis-generation → validation project – summary for the driver-negative LUAD investigator

1. Why we did this

Oncogene-negative ("driver-negative", RTK/RAS/RAF pathway wild-type) LUAD remains the enigmatic tail of the disease: ~15–20% of adenocarcinomas with no actionable driver, poorly served by targeted therapy and variably by immunotherapy. Our entry point was a mechanistic hypothesis generated by the **Asta Theorizer**: that in these tumors **chromosomal instability (CIN) is the oncogenic engine, and that CIN drives immune escape** – with the second-generation checkpoints **TIGIT and LAG3** as the proposed effectors (panel-aware against PD-1/PD-L1/TIM-3/CTLA-4).

The question we set out to answer: *is that mechanism real and specific to driver-negative LUAD, and does it point to a therapeutic axis (e.g. LAG3 blockade) for a group that badly needs one?*

Starting data (patient-provided): a TCGA-LUAD multi-omics set — gene-level copy number, RNA-seq, somatic mutations (MAF), TMB, MSigDB Hallmark ssGSEA scores, and an **RPA (RTK/RAS/RAF Pathway Alteration) classification** giving 118 driver-negative vs 383 driver-positive cases.

2. Methodology and the Asta agents used

We ran a full **generate** → **stress-test** → **validate** loop, deliberately trying to *break* each positive finding rather than confirm it.

ASTA AGENT	ROLE IN THIS PROJECT
Theorizer	Generated the founding CIN → immune-escape → TIGIT/LAG3 hypothesis.
Paper Finder (<code>asta literature find</code>)	8 literature searches — a structured gap analysis (615 unique papers) + mechanism/biology grounding. Established that the <i>specific</i> junction (CIN ↔ TIGIT/LAG3 in driver-negative LUAD) was an unstudied white space.
AutoDiscovery (AutoDS)	6 runs / 120 autonomous experiments — Bayesian-surprise-guided hypothesis testing across the five gaps (association, driver-specificity, checkpoint selectivity, cGAS-STING mediation, specific CN events), plus a decisive infiltration-normalized follow-up.
DataVoyager (<code>asta analyze-data</code>)	5 targeted, code-executing validation analyses (E1, E2, E1b, E7-interaction) that re-tested the AutoDS leads under gold-standard metrics.

We supplemented Asta with public reference data and a custom single-cell pipeline: - **GDC PanCanAtlas** — Thorsson *leukocyte fraction* + CIBERSORT (independent CD8 abundance), Taylor **Aneuploidy Score** (purity-aware CIN), ABSOLUTE **purity/ploidy/WGD**; **TIMER2.0** (6-method CD8). - **LuCA** core single-cell atlas (Salcher 2022; 892k cells) and a cohort-agnostic inferCNV → CD8-pseudobulk → tumor-level pipeline we built and unit-tested.

A methodological principle we enforced throughout: the independent unit is the **tumor**, not the cell or the gene. Single-cell inference was done at tumor/patient level (pseudobulk), and we treated tumor purity, CIN-metric choice, and infiltration quantification as first-class confounders.

3. Avenues tested, and what each returned

STAGE	QUESTION	TOOL	RESULT
Literature	Is the CIN → TIGIT/LAG3 link in driver-negative LUAD studied?	Paper Finder	Genuine white space (0 papers joining CIN + driver-negative; 2 generic CIN↯TIGIT/LAG3).
AutoDS G1	Does CIN associate with TIGIT/LAG3?	AutoDiscovery	Bulk association is negative – CIN tumors are checkpoint- low . Original "up-regulation" direction refuted.
AutoDS G2	Is it driver-negative-specific?	AutoDiscovery	No – same in driver-positive (no interaction).
AutoDS G3	Selective for TIGIT/LAG3 vs PD-1/PD-L1?	AutoDiscovery	No selectivity.
AutoDS G4	Mediated by cGAS-STING/IFN?	AutoDiscovery	Not mediated (CIN suppresses STING, but no formal mediation).
AutoDS G5	Which CN events?	AutoDiscovery	Global CIN + STK11 loss → immune-cold; <i>not</i> 9p21/CDKN2A.
AutoDS run6	Per <i>T cell</i> (adjust for infiltration)?	AutoDiscovery	Apparent rescue : CIN → fewer T cells, but per-cell LAG3 ↑ ($\beta=2.81$, $p=0.001$) , TIGIT null, IFN- γ ↑. LAG3-specific.
E1	Holds under gold-standard metrics (Aneuploidy Score, absolute CD8, purity)?	DataVoyager	No – LAG3~CIN $p=0.85$; CIN ↔ infiltration $r=-0.03$.
E1b	Proxy vs Aneuploidy Score, $\pm$ purity, dissected	DataVoyager	Infiltration-exclusion = purity artifact (proxy $r=0.54$ with purity). Per-cell LAG3 is purity-robust & LAG3-specific but covariate-dependent (significant only with an expression-derived T-cell score).
E7	Does driver status modify CIN → LAG3 (formal interaction)?	DataVoyager	Weak/absent – effect is general, not driver-negative-specific.
Feasibility	Can single-cell test the driver-negative claim?	LuCA audit	No – only 1 eligible onco-negative primary tumor.
LuCA	Does per-cell LAG3~CIN replicate in single cell (all-LUAD)?	custom inferCNV pipeline	Null – LAG3~CIN $\beta=0.026$, $p=0.56$ across 43 primary tumors.

4. What we learned

About the biology of driver-negative LUAD

1. **The founding hypothesis is not supported.** CIN does **not** drive TIGIT/LAG3-mediated immune escape in driver-negative LUAD. The most enticing lead – a per-cell LAG3 up-regulation – was covariate-dependent in bulk and **did not replicate** when CD8 LAG3 was measured directly in single-cell data ($\beta \approx 0$). LAG3 blockade does not gain a rationale here from this dataset.
2. **Driver-negative LUAD is not a distinct CIN-immune compartment.** Whatever CIN does to the immune microenvironment, it does about equally in oncogene-positive tumors (no interaction). Framing CIN immunobiology as *specific* to the driver-negative subset is not warranted.
3. **What CIN association survives is immune "coldness," and much of even that is tumor purity.** The apparent CIN $\rightarrow$ reduced T-cell infiltration signal collapsed once we used a purity-corrected aneuploidy score and adjusted for ABSOLUTE purity. The one durable genomic correlate of coldness was **STK11 loss** – consistent with, not novel beyond, the known STK11/KEAP1 cold-tumor axis.

Methodological lessons (these are the most transferable findings)

4. **CIN-metric choice changes the answer.** An expression/amplitude "fraction-genome-altered" proxy and the arm-level **Taylor Aneuploidy Score** are only moderately correlated ($r \approx 0.6$) and gave *opposite* significance. Bulk CIN-immunology conclusions built on FGA-type proxies should be treated with suspicion until reproduced with a purity-aware aneuploidy score.
5. **Tumor purity is a first-order confounder** in any bulk CIN $\leftrightarrow$ immune analysis (it inflates proxy-CIN *and* dilutes the RNA immune signal). It must be modeled explicitly.
6. **"Per-cell" claims from bulk are fragile.** Adjusting a checkpoint for an expression-derived infiltration score (built from the same RNA) induces residual co-expression that can masquerade as a per-cell effect. Deconvolution disagreed with regression adjustment; single cell settled it.

Practical / feasibility

7. **Driver-annotated single-cell LUAD does not exist at usable scale.** LuCA (the largest atlas) has driver annotation for only ~75/161 LUAD patients and just **one** eligible oncogene-negative primary tumor. Any single-cell interrogation of a *driver-negative-specific* mechanism is currently infeasible with public data – a real constraint for our subfield.

5. Implications for the driver-negative LUAD research program

- **Don't build a LAG3 (or TIGIT) therapeutic hypothesis for oncogene-negative LUAD on TCGA bulk checkpoint expression.** It doesn't survive rigorous re-analysis or single-cell validation.
- **Re-audit prior bulk CIN-immune claims** in this space with purity-corrected aneuploidy metrics and, where possible, single-cell or deconvolution – several may be purity/metric artifacts.
- **The interesting, still-open threads** are the ones that survived: (a) *what actually drives* oncogene-negative tumorigenesis (tumor-suppressor-loss/CIN biology), and (b) the **STK11-loss cold program** as a concrete, better-supported immune-evasion lead worth pursuing on its own.

- If a CIN → LAG3 question is ever revisited, it needs a driver-annotated single-cell or spatial cohort with LAG3 protein per T cell — data that would have to be generated, not mined.

6. What worked about the approach

The Asta loop was effective precisely because it produced a *falsifiable* lead fast (Theorizer → AutoDS surfaced the per-cell LAG3 signal in hours) and then let us **kill it responsibly**: DataVoyager re-tests exposed the metric/purity/covariate fragility, and the single-cell pipeline delivered the clean negative. The headline deliverable is not a new drug target — it's a well-documented negative result plus a set of methodological guardrails for bulk CIN-immunology, which is worth more than a false-positive lead pursued into the wet lab.

7. Where the artifacts live (`charliemcgrady/research` , branch `claudio/asta-plugins-install-qi0zqc`)

- `asta_lit/GAP_ANALYSIS.md` — literature gap analysis (Paper Finder).
- `asta_autods_data/runs/SYNTHESIS.md` + `RUNS.md` — the 6 AutoDiscovery runs, configs, per-experiment JSON.
- `lag3_validation/` — the full validation program:
- `EVIDENCE_CATALOG.md` , `DATAVOYAGER_EXPERIMENTS.md` — data sources and the 10-experiment plan.
- `TIER1_RESULTS.md` , `E1B_RESULTS.md` , `E7_INTERACTION_RESULTS.md` — DataVoyager validation.
- `SC_FEASIBILITY_AUDIT.md` — LuCA driver-annotation audit.
- `LUCA_GENERAL_RESULTS.md` — the decisive single-cell null.
- `pipeline/` — cohort-agnostic single-cell pipeline (+ passing synthetic smoke test).
- `datasets/` , `scrna/luca/general_out/` — analysis-ready tables and results.

Workflow: Theorizer → Paper Finder → AutoDiscovery → DataVoyager, with a single-cell coda outside the agents. Documents are the versions committed to the repository. Styled with the Ai2 Asta design system; Manrope and Roboto Mono embedded, PP Telegraf falls back to Manrope.